Wisdom of Crowds with Many Separate Groups: A Correction to Golub and Jackson (2010)

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Abstract

Golub and Jackson (2010) show that naive DeGroot learning makes a large society wise if and only if the most influential agent's influence vanishes (their Proposition 2), and state that their results apply to all convergent networks, not only strongly connected ones. For networks with a growing number of separate closed groups, the "if" direction does not hold under their definition of wisdom, which requires all agents to be close to the truth simultaneously. A network of 4^m groups of m agents satisfies every assumption and the influence condition, yet with probability $1-(1-2^{-m})^{4^m} \to 1$ some agent ends at belief one when the truth is one half. Theorem 4, whose proof starts from Proposition 2, fails on the same network. The "only if" direction, Proposition 2 itself when the number of groups is bounded (including every strongly connected network), and the paper's central message are unaffected. We give a corrected statement that can replace Proposition 2, a sharp threshold for equally weighted groups (group size must grow faster than the logarithm of the number of groups), and show that Theorem 1 is not exposed to the problem.

1 The statements in question

Society n has a row-stochastic matrix $T(n)$ and beliefs $p(t) = T(n)^t p(0)$. Initial signals $p_i(0)$ are independent, lie in $[0,1]$, have common mean μ, and have variances bounded below by some $v > 0$. When $T(n)^t$ converges, let $C_1, \dots, C_K$, $K = K_n$, be its minimal closed classes ("closed groups"). Each class C_c has a stationary probability vector w_c, and its members converge to the class consensus $Y_c = \sum_{i \in C_c} w_{ci}\, p_i(0)$; other agents converge to averages of the Y_c, so

$$\max_{i \le n} |p_i(\infty) - \mu| = \max_{c \le K_n} |Y_c - \mu|. \tag{1}$$

Let $a_n = \max_{c,i} w_{ci}$ denote the largest influence; this is the paper's $s_1(n)$.

(a) **Definition 3** calls $(T(n))$ *wise* if $\max_{i \le n} |p_i(\infty) - \mu| \to 0$ in probability.

(b) **Proposition 2** states that a sequence of convergent stochastic matrices is wise if and only if $s_1(n) \to 0$.

(c) **Footnote 11** states that strong connectedness is not required: "All the results go through for general convergent networks."

(d) **Theorem 4** states that a sequence of convergent stochastic matrices satisfying balance (Property 1) and minimal in-dispersion (Property 3) is wise. Its proof reduces the claim to $s_1(n) \to 0$ by Proposition 2.

2 Counterexample

Counterexample. *For $m \ge 2$, let society $n = m\,4^m$ consist of 4^m disjoint groups of m agents, each agent putting weight $1/m$ on every member of its own group (including itself) and zero weight outside it. Let signals be independent fair coins, $p_i(0) \in \{0,1\}$, so $\mu = \frac{1}{2}$. Then:*

(i) every assumption of the paper holds and $T(n)$ is convergent;

(ii) the largest influence is $s_1(n) = 1/m \to 0$;

(iii) $\Pr\big(\max_i |p_i(\infty) - \frac{1}{2}| = \frac{1}{2}\big) = 1-(1-2^{1-m})^{4^m} \ge 1-(1-2^{-m})^{4^m} \ge 1-e^{-2^m} \to 1$, where the middle term is the probability that some agent ends at belief one.

Hence the sequence satisfies the condition of Proposition 2 and the hypotheses of Theorem 4 (Section 4) but is not wise.

Proof. (i) $T(n)$ is nonnegative and row-stochastic. Each block equals J_m/m, where J_m is the $m \times m$ matrix of ones; $(J_m/m)^2 = J_m/m$, so $T(n)^t = T(n)$ for all $t \geq 1$ and beliefs converge after one step. Fair coins are independent, lie in $[0,1]$, have mean $\frac{1}{2}$ and variance $\frac{1}{4} > 0$. (ii) Each group is a closed class with uniform stationary vector, so every influence equals $1/m$. (iii) By (1) the maximum error equals $\frac{1}{2}$ exactly when some group average is 0 or 1. Groups are independent and each is all zeros or all ones with probability 2^{1-m}, which gives the first expression; the event that some group is all ones has probability $1-(1-2^{-m})^{4^m}$, which is smaller. Finally $(1-2^{-m})^{4^m} \leq \exp(-4^m 2^{-m}) = e^{-2^m}$. □

m	groups	agents n	largest influence	Pr(some agent ends at belief 1)
2	16	32	1/2	0.98998
3	64	192	1/3	0.99981
4	256	1,024	1/4	$1 - 6.7 \times 10^{-8}$
5	1,024	5,120	1/5	$1 - 7.6 \times 10^{-15}$

Table 1: Exact probabilities for the Counterexample; the appendix gives a short check.

Remark. The counterexample does not depend on special population sizes. For every $n \geq 16$, groups of size $\lfloor \frac{1}{2} \log_2 n \rfloor$ or one more give largest influence of order $1/\log n$ and a probability of failure at least $1 - \exp(-K_n/(2\sqrt{n})) \to 1$, because the number of groups K_n grows faster than $\sqrt{n}$. Nor does it depend on coin signals: if every signal satisfies $\Pr(p_i(0) \geq \mu + \delta) \geq \eta$ for some $\delta, \eta > 0$, groups of size $m \to \infty$ fail in the same way whenever their number grows faster than η^{-m}, since a group whose members all exceed $\mu + \delta$ has consensus at least $\mu + \delta$.

3 Where the proof of Proposition 2 slips

The proof reduces the claim to showing that $p_{i_n}(\infty) \to \mu$ in probability for an arbitrary sequence of agents i_n in closed groups, and establishes that by applying the weighted law of large numbers (Lemma 1) to the group of i_n, using its own influence weights. This argument is correct and shows

$$\sup_{i \leq n} \Pr\big(|p_i(\infty) - \mu| > \varepsilon\big) \to 0.$$

Definition 3 requires

$$\Pr\Big(\max_{i \leq n} |p_i(\infty) - \mu| > \varepsilon\Big) \to 0.$$

The first statement concerns one agent at a time; the second concerns all agents simultaneously. The agent with the largest error depends on the realized signals, so no deterministic sequence i_n can track it. By (1) and independence across groups, the two statements agree when K_n is bounded, and differ when K_n grows fast enough relative to the size of the groups.

The Counterexample shows the gap explicitly. Take any deterministic sequence of agents i_n. The limiting belief of i_n is the average of m independent fair coins, so by Hoeffding's inequality

$$\Pr\big(|p_{i_n}(\infty) - \tfrac{1}{2}| > \varepsilon\big) \leq 2e^{-2m\varepsilon^2} \to 0,$$

uniformly over the choice of i_n. The step in the proof therefore succeeds for every such sequence, exactly as written. Yet $\Pr(\max_i |p_i(\infty) - \frac{1}{2}| = \frac{1}{2}) \to 1$. Each group fails with small probability, but there are 4^m independent groups; the union bound $4^m \cdot 2e^{-2m\varepsilon^2} = 2e^{m(\log 4 - 2\varepsilon^2)}$ diverges for small ε, and the exact calculation in the Counterexample shows that some group does fail with probability tending to one.

4 Theorem 4

Balance (Property 1) requires a sequence $j(n) \to \infty$ such that, for families B_n with $|B_n| \leq j(n)$, the ratio of the weight B_n receives from the rest of society to the weight it gives to the rest of society stays bounded. Minimal in-dispersion (Property 3) requires $q \in \mathbb{N}$ and $r < 1$ such that, for $|B_n| = q$ and every finite family $C_n \subseteq B_n^c$, the weight C_n gives to B_n is at most r times the weight B_n gives to B_n^c, for all large n. As in the paper's use of the property, a finite family means a sequence of sets whose cardinality is bounded uniformly in n.

In the network of the Counterexample, take $j(n) = m - 1 \to \infty$. A family of at most $m-1$ agents is a proper subset of every group it meets; by symmetry of each block it gives and receives the same, strictly positive, weight, so the ratio equals one and no 0/0 arises. Take $q = 1$ and $r = \frac{1}{2}$: a single agent gives weight $1 - 1/m$ to the rest of society, while a set of at most h other agents gives it at most h/m, which is below $\frac{1}{2}(1 - 1/m)$ once $m > 2h + 1$. Both properties hold, the matrices are convergent, and the sequence is not wise.

5 A corrected statement

Write $Q_c = \sum_{i \in C_c} w_{ci}^2$ for the concentration of influence within closed group c.

Corrected Proposition 2. *Let $(T(n))$ be convergent stochastic matrices, with signals as above, and let $Y_1, \ldots, Y_{K_n}$ be the consensuses of the closed groups.*

(i) The sequence is wise if and only if, for every $\varepsilon > 0$,

$$\sum_{c=1}^{K_n} \Pr\big(|Y_c - \mu| > \varepsilon\big) \longrightarrow 0.$$

(ii) If the sequence is wise, then $s_1(n) \to 0$.

(iii) If the number of closed groups K_n is bounded, the sequence is wise if and only if $s_1(n) \to 0$. This includes all strongly connected networks ($K_n = 1$).

Proof. (i) With $\pi_c = \Pr(|Y_c - \mu| > \varepsilon)$, independence across groups and (1) give $\Pr(\max_i |p_i(\infty) - \mu| > \varepsilon) = 1 - \prod_c (1 - \pi_c)$, which lies between $1 - e^{-\sum_c \pi_c}$ and $\sum_c \pi_c$. (ii) If $s_1(n) \geq \delta$ along a subsequence, the corresponding Y_c has variance at least $v\delta^2$; since $|Y_c - \mu| \leq 1$, $\Pr(|Y_c - \mu| > \varepsilon) \geq v\delta^2 - \varepsilon^2 > 0$ for small ε, so the sum in (i) does not vanish. (iii) Necessity is (ii). For sufficiency, $\mathrm{Var}(Y_c) \leq Q_c/4 \leq s_1(n)/4$, so by Chebyshev $\sum_c \pi_c \leq K_n \, s_1(n)/(4\varepsilon^2) \to 0$. □

Corollary (A distribution-free sufficient condition)**.** *If $\big(\max_c Q_c\big)\big(1 + \log K_n\big) \to 0$, the sequence is wise.*

Proof. Hoeffding's inequality for weighted sums of independent variables in $[0,1]$ gives $\pi_c \leq 2\exp(-2\varepsilon^2/Q_c)$, so $\sum_c \pi_c \leq 2\exp(\log K_n - 2\varepsilon^2/\max_c Q_c)$. The condition implies both $\max_c Q_c \to 0$ and $(\max_c Q_c)\log K_n \to 0$, so the exponent, $-(2\varepsilon^2 - \max_c Q_c \log K_n)/\max_c Q_c$, tends to $-\infty$, and part (i) applies. The term 1 matters: with a single group, $\log K_n = 0$, and $\max_c Q_c \to 0$ must still be required. □

Remark. Vanishing maximal influence, $s_1(n) \to 0$, is equivalent to individual learning, $\sup_i \Pr(|p_i(\infty) - \mu| > \varepsilon) \to 0$ for every $\varepsilon > 0$: one direction is the Chebyshev bound in the proof of (iii) applied to a single group (and to averages of group consensuses, whose variance is no larger, for agents outside closed groups), the other is the argument in (ii). This is what the original proof establishes.

For Theorem 4, the same correction applies to its conclusion: wisdom follows if, in addition, K_n is bounded or the condition of the Corollary holds. We have not re-examined the remaining steps of its proof, which establish $s_1(n) \to 0$.

In the Counterexample, $\max_c Q_c \, (1 + \log K_n) = (1 + m \log 4)/m \to \log 4$, so the Corollary's condition fails, as it must. For equally weighted groups the condition is also necessary, in the following sense.

Proposition (a sharp threshold). *Suppose society n consists of K_n disjoint closed groups of common size m_n, each agent putting weight $1/m_n$ on every member of its own group, with independent fair-coin signals. The sequence is wise if and only if $m_n \to \infty$ and $(\log K_n)/m_n \to 0$.*

Proof. Write $m = m_n$, $K = K_n$ and $I(x) = (\frac{1}{2} + x)\log(1 + 2x) + (\frac{1}{2} - x)\log(1 - 2x)$, the Kullback–Leibler divergence of a coin with mean $\frac{1}{2} + x$ from a fair coin; I is continuous and increasing on $[0, \frac{1}{2})$ with $I(0) = 0$. Sufficiency: by Hoeffding's inequality, $\sum_c \pi_c \leq 2Ke^{-2m\varepsilon^2} = 2\exp\{m(\log K/m - 2\varepsilon^2)\} \to 0$, and part (i) applies. Necessity: if $m_n \not\to \infty$, then $s_1(n) = 1/m_n \not\to 0$ and part (ii) applies. Otherwise suppose $\log K/m \geq c > 0$ along a subsequence with $m \to \infty$. Fix $\delta > 0$ with $I(\delta) < c$ and let $k = \lceil m(\frac{1}{2} + \frac{\delta}{2}) \rceil$, so that $\frac{1}{2} + \frac{\delta}{2} \leq k/m \leq \frac{1}{2} + \delta$ for large m. The method-of-types bound (Cover and Thomas, 2006, Theorem 11.1.4) gives

$$\Pr\big(Y_c \geq \tfrac{1}{2} + \tfrac{\delta}{2}\big) \geq \binom{m}{k} 2^{-m} \geq \frac{e^{-mI(k/m - 1/2)}}{m+1} \geq \frac{e^{-mI(\delta)}}{m+1}.$$

With $\varepsilon = \delta/4$, $\sum_c \pi_c \geq \exp\{m(c - I(\delta)) - \log(m+1)\} \to \infty$, so by part (i) the sequence is not wise. □

The threshold has a simple reading: the number of signals in each group must grow faster than the logarithm of the number of groups. In the Counterexample, $\log K_n/m_n = \log 4$ stays positive.

Remark (Average and worst-case learning)**.** Vanishing maximal influence does make the society accurate on average: every limiting belief has variance at most $s_1(n)/4$, so $E\big[\frac{1}{n}\sum_i (p_i(\infty)-\mu)^2\big] \le s_1(n)/4 \to 0$. In the Counterexample this average equals $1/(4m) \to 0$, while $\max_i |p_i(\infty) - \frac{1}{2}| \to \frac{1}{2}$ in probability. The original influence condition is therefore the right condition for learning by almost everyone; Definition 3 asks that nobody be substantially wrong, and that requires the additional condition in part (i).

6 Theorem 1

Theorem 1 states that balance and minimal out-dispersion (Property 2) imply wisdom. Minimal out-dispersion requires $q \in \mathbb{N}$ and $r > 0$ such that, whenever B_n is a finite family with $|B_n| \ge q$ and $|C_n|/n \to 1$, the weight B_n gives to C_n exceeds r for all large n. This property bounds the number of closed groups. If K_n were unbounded, then along a subsequence with $K_n \to \infty$ the union U_n of the q smallest closed groups would satisfy $|U_n|/n \le q/K_n \to 0$. Taking B_n to be one agent from each of these groups and C_n the complement of U_n gives $|B_n| = q$ and $|C_n|/n \to 1$, yet B_n gives no weight outside U_n, so the weight B_n gives to C_n is zero, a contradiction. Hence the obstruction identified here does not arise under Theorem 1's hypotheses, and by part (iii) of the Corrected Proposition, Theorem 1 holds provided its proof that $s_1(n) \to 0$ is valid. We have not re-examined that part of the proof.

7 What is unaffected

The "only if" direction of Proposition 2; the full equivalence of Proposition 2 when the number of closed groups is bounded, including every strongly connected network (wisdom still requires $s_1(n) \to 0$ there); Corollary 1 for connected undirected networks; Theorem 1, whose out-dispersion condition rules out many closed groups (Section 6); the obstruction created by prominent families (Proposition 3); the examples; and the paper's central insight that a society learns well only if no agent's influence stays large. The correction concerns the general, not strongly connected, case announced in footnote 11. We have not established whether this observation appears elsewhere.

Appendix: exact check

The following standard-library Python reproduces the table.

```
from fractions import Fraction
from math import exp
for m in range(2, 9):
    groups = 4 ** m
    p_fail = 1 - (1 - Fraction(1, 2 ** m)) ** groups
    print(m, m * groups, f"1/{m}", float(p_fail), 1 - exp(-2 ** m))
```

References

[1] Cover, T. M., and Thomas, J. A. (2006). *Elements of Information Theory*, 2nd ed. Hoboken, NJ: Wiley.

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[3] Hoeffding, W. (1963). Probability Inequalities for Sums of Bounded Random Variables. *Journal of the American Statistical Association* 58(301), 13–30.

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